

03 BioColor Tag

1 Background

Scale of Food Waste

According to FAO data, approximately 1.3 billion tons of food are wasted globally each year, with perishable foods accounting for the highest proportion.

FRESH
INDICATOR

Overview

Freshness indicator label based on natural colorings

Theme

Nature-Driven Sensing
Food Safety & Sustainability
Design for Visual Interaction



Management Burden

For families, managing perishable foods is a burden: it requires constantly monitoring food conditions and accurately determining their edibility and shelf life, a process that often causes psychological stress and operational inconvenience.



3 Summary

Existing "time labels" and "expensive electronic monitoring" failed to meet the mass-market demand for low-cost, intuitive freshness detection; consumers struggle to make accurate judgments, and retailers lack visual display tools, resulting in food waste and trust crises coexisting.

Economic Losses

Food waste results in global economic losses of approximately \$940 billion per year.

Cold Chain Failure

Approximately over 25% of perishable foods spoil due to improper temperature control in the coldchain.

Consumer Misjudgment

According to an NRDC survey, 91% of consumers misunderstand "best before" dates and discard food that is still edible.

2 Pain Point Interviews



Jenny Wu

Hotel Manager

Needs a reliable way to ensure buffet and stored ingredients stay fresh, since staff often misjudge food quality.



Feng

Independent Artist

Lives alone and forgets what's in the fridge leading to food spoilage or unsafe eating.



Jia Qi

Fund Manager

Buys premium food but discards it too early out of uncertainty, wasting money and nutrition.



Yu dong

Clothing Agent

Travels often and returns unsure if stored meat or dairy is still safe to eat.



Supermarkets and fresh food retailers lack intuitive freshness display tools in shelf management.



A UK WRAP report indicates that approximately 50% of household food waste stems from misjudging freshness, particularly for meat and dairy products.

4 Color Reaction Experiment

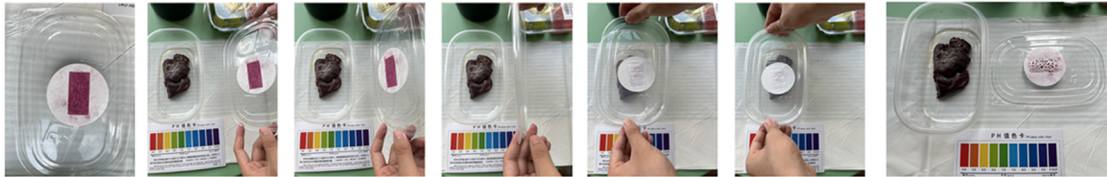
First, the study tested anthocyanin and curcumin under the form of dry powder-observing their color changes based on pH response.



Curcumin powder color change



Anthocyanin powder color change



Conclusion - 1

Experiment turned out as expected, dry powders are chemically stable, color changes are extremely slow and nearly invisible, thus not applicable.

Anthocyanin Liquid Extraction Process



Select purple cabbage



Cut into small pieces

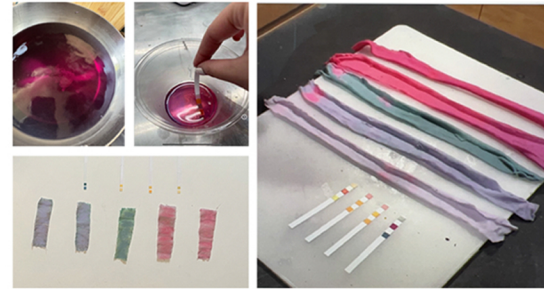


Boil to create anthocyanin solution

5 Material Adhesion Experiment

In adhesion experiments on paper and fabric substrates, anthocyanins demonstrated excellent coloring performance.

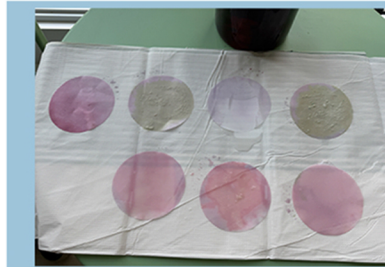
However, when considering the feasibility of subsequent industrial applications, **paper-based materials** showed more significant advantages.



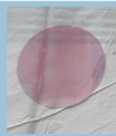
Conclusion - 2

Anthocyanin extract mixed with three acids and three alkalis showed significant color changes within five minutes. Results from mixing hydrogen sulfide, ammonia, and other gases with volatile compounds. This process also warns us that meat is already rich in microorganisms, and the gas mixture produced by rotting meat contains high concentrations of ammonia.

PH test experiment



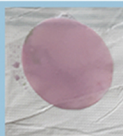
Acid



White vinegar

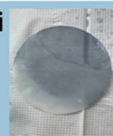


lemon water

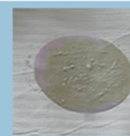


hair conditioner

Alkali



Shampoo



laundry detergent



baking soda

6 Meat As Reaction Testing Subject

Step1

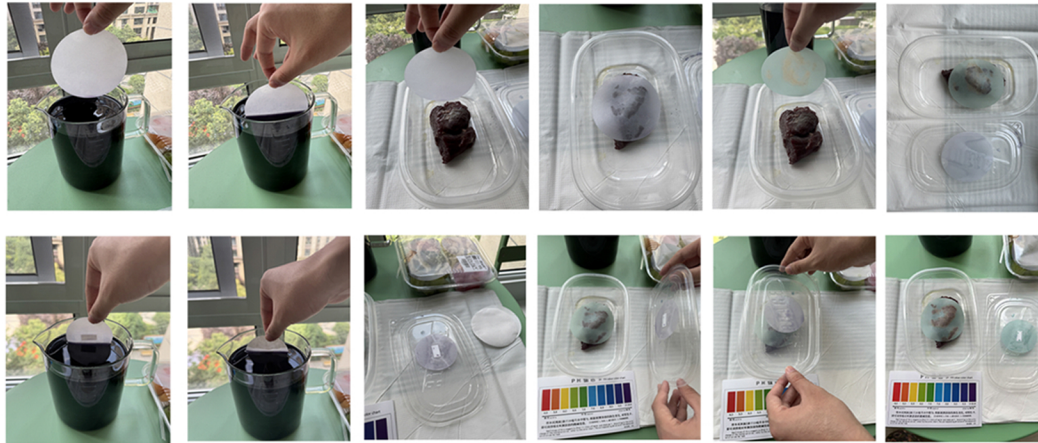
Extract anthocyanin aqueous solution

Step2

Start experiment, corresponding to Figure 3, put the test paper on the meat.

Step3

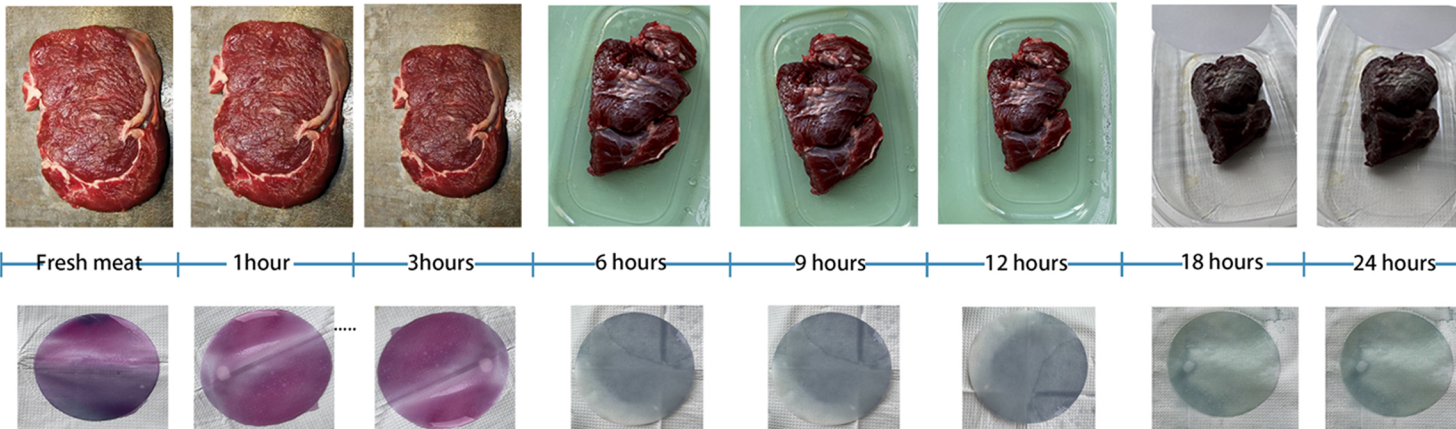
Corresponding to Figure 5, the test paper has undergone a color change reaction.



Observation

Anthocyanins, as water soluble natural pigments, exhibit relatively stable chemical properties in powder form but demonstrate high reactivity in aqueous solutions. Therefore, to utilize anthocyanins for detecting decaying gases, they must be maintained in a moist or solution state.

Meat Decomposition Process



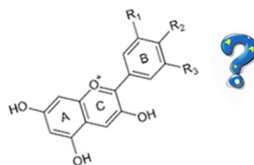
Conclusion

At room temperature, the freshness quality deterioration of beef during the post-mortem phase is indicated by monitoring changes in anthocyanin color.

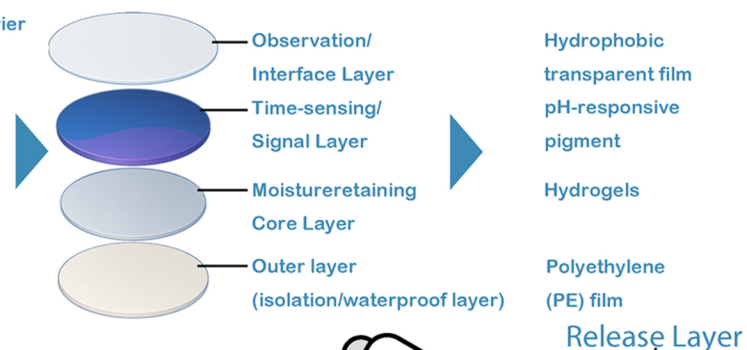
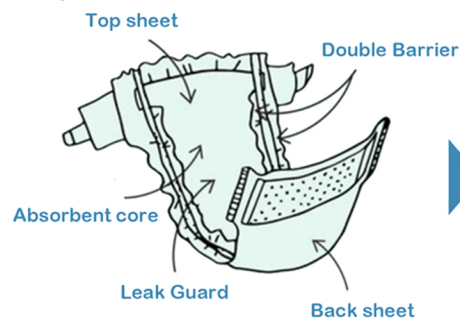
7 Product Design Process

Conclusions and future research directions
The experimental results indicate that when using anthocyanins to detect spoilage gases maintaining them in a moist or solution state is a necessary prerequisite.

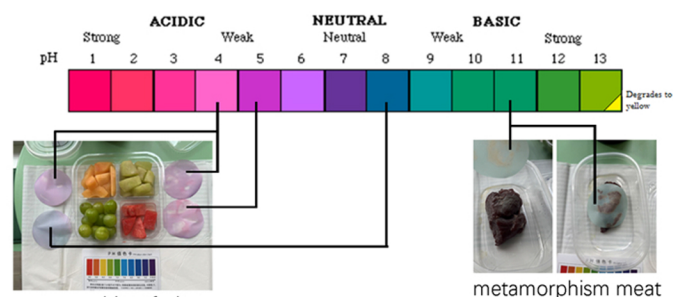
Therefore, how to effectively maintain the appropriate humidity environment (moisture retention) of anthocyanin sensor units/materials has become a key technical issue that needs to be addressed to realize their practical application.



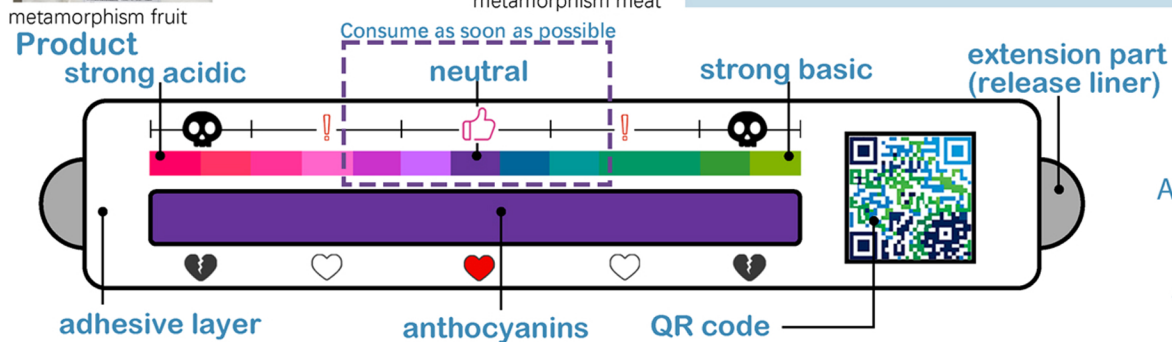
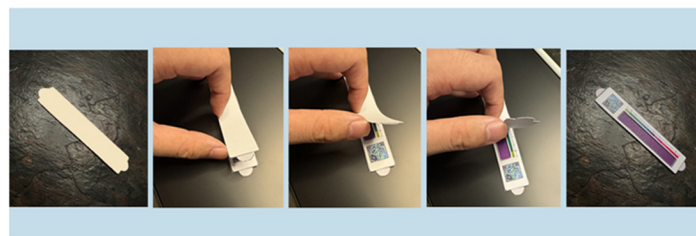
8 Inspiration



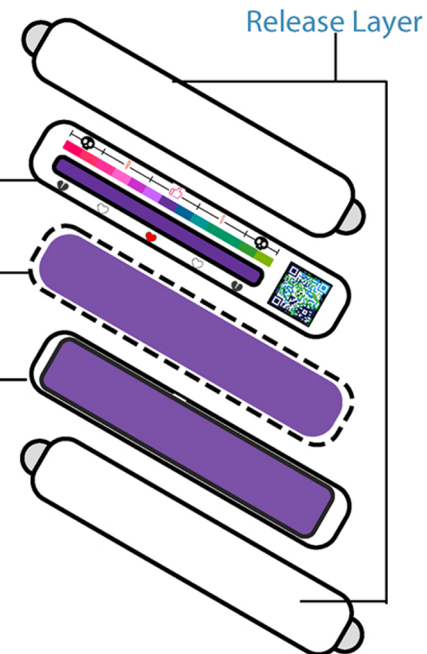
Freshness Indicator



Adhesive Layer Tearing Process



Visible window
Anthocyanin Absorbent Core Layer
Anthocyanin Moisture-Lock Layer



9 Interface



10 Scenarios

A boy came home and reached for strawberries in the fridge. The red label had turned dark-his mother stopped him: "They're spoiled." He realized the tiny label's color was protecting his health.



Future

Future anthocyanin tags may introduce a new pricing indicator for supermarkets.



Innovative



Eco-Friendly



Practical Application

Through interdisciplinary design and systematic verification, this product will provide new insights for smart packaging and biosensing fields, demonstrates the application potential of natural molecules in green sustainable technologies, and promotes their transition from laboratory to practical application.

Consume as soon as possible

No Longer Edible

Consume as soon as possible

